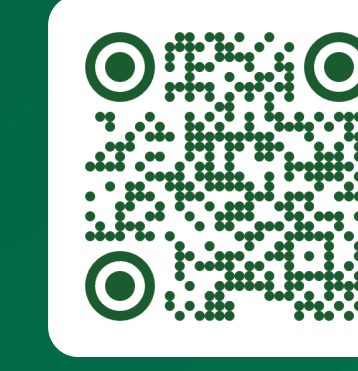


DisentangledTMR: Privacy-Preserving Skeleton Motion Retargeting via Factorized Transformers

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PRIVACY GAIN

75.4 → 18.1 %

Re-identification falls from **30× chance** to **7.2×**, against a 2.5% floor

UTILITY KEPT

75.8%

Action recognition with **no retraining**, and 87.1% once downstream models are retrained

VS. PMR

2.1 × utility

75.8% against PMR's 35.7% under the same protocol, at a comparable privacy level

TUNABLE

$\beta = 0.2$

One knob sets the operating point after training: **76.8% AR at 17.3% RI**

The problem

3D skeletons look anonymous, but bone lengths, limb proportions and gait are a **biometric signature**: an off-the-shelf recogniser re-identifies the actor in raw NTU skeletons **75.4% of the time** across 40 identities, 30× chance.

Yet skeletons are exactly the modality we want to share for health monitoring, elder care and behaviour analysis. Blurring or adding noise destroys the action along with the identity.

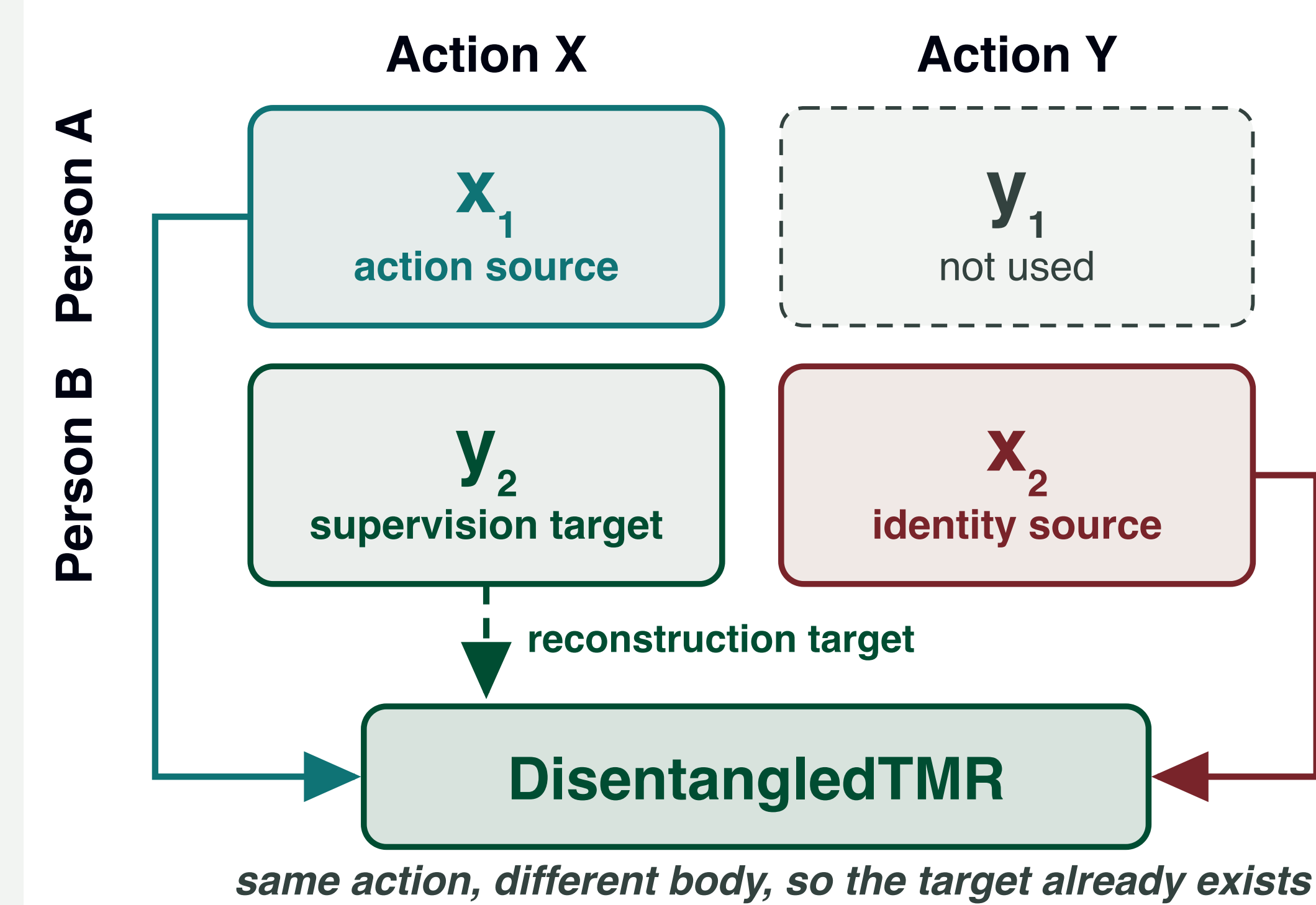
The idea

Replace identity rather than removing it. Encode the motion of one person and the body of another into two separate latent streams, then decode a sequence that performs the **source action** on the **target skeleton**.

- **Action stream** (768-d) captures dynamics: position, velocity, acceleration → multi-scale temporal convolution → attention.
- **Identity stream** (256-d) captures structure: mean static pose + bone lengths → spatial GCN. Deliberately **low-capacity**, an information bottleneck that starves it of motion detail.
- **Asymmetric by design**. The architecture does most of the disentangling; the contrastive, adversarial, orthogonality and cross-correlation losses do the rest.

Training signal

Every training sample is a **cross-identity quadruplet**: two actors who each perform the same two actions, so the exact retargeting target already exists in the data and the reconstruction term needs no adversarial proxy.



Three-stage training

Order is mandatory. Skipping a stage measurably degrades disentanglement.

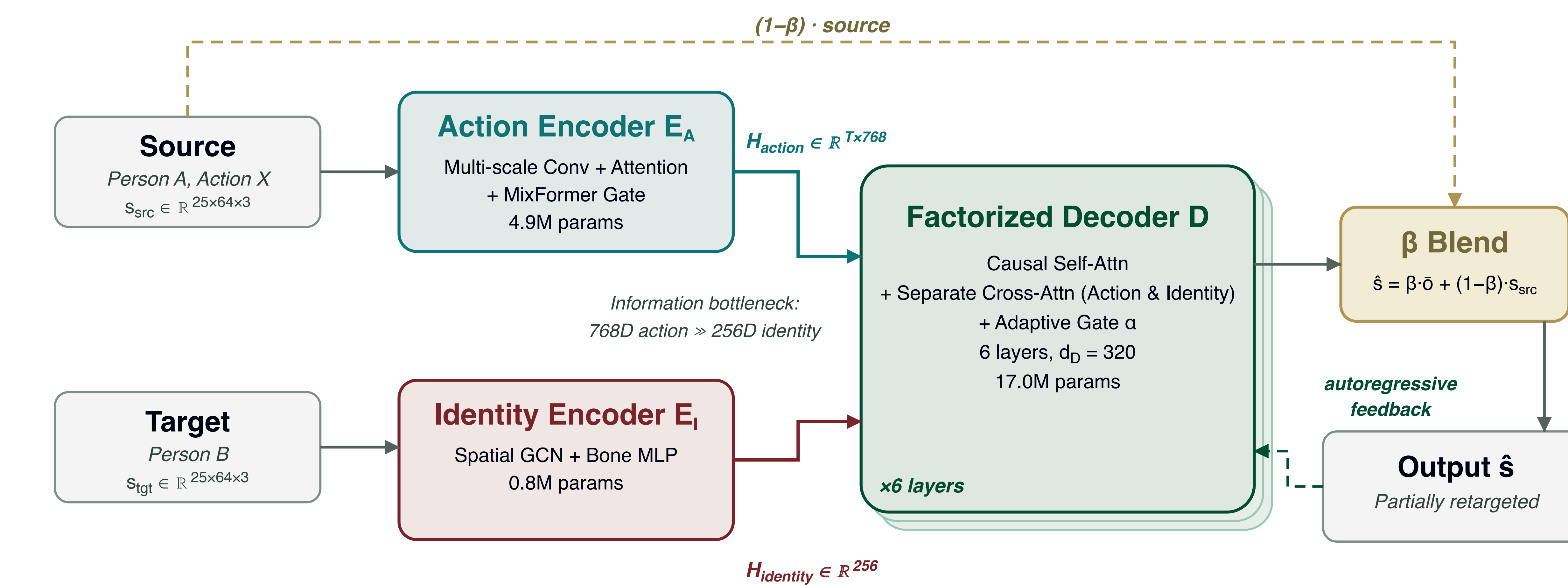
- 1 **Encoder pre-training**. Both encoders learn with classification heads plus the four disentanglement losses. Decoder frozen.
- 2 **Decoder training**. Encoders frozen. Reconstruction plus physical plausibility (bone length, smoothness, velocity, end-effector, foot contact, joint limits). Teacher forcing 1.0 → 0.5.
- 3 **End-to-end fine-tuning**. All components jointly, teacher forcing 0.5 → 0.3, with output-level adversarial and cooperative heads.

Scope

Data. NTU RGB+D 60 / 120 and ETRI-Activity3D, single-person sequences only (two-person NTU actions excluded).

Privacy is measured against trained recognisers, not a worst-case adversary with full knowledge of the retargeting model.

Architecture

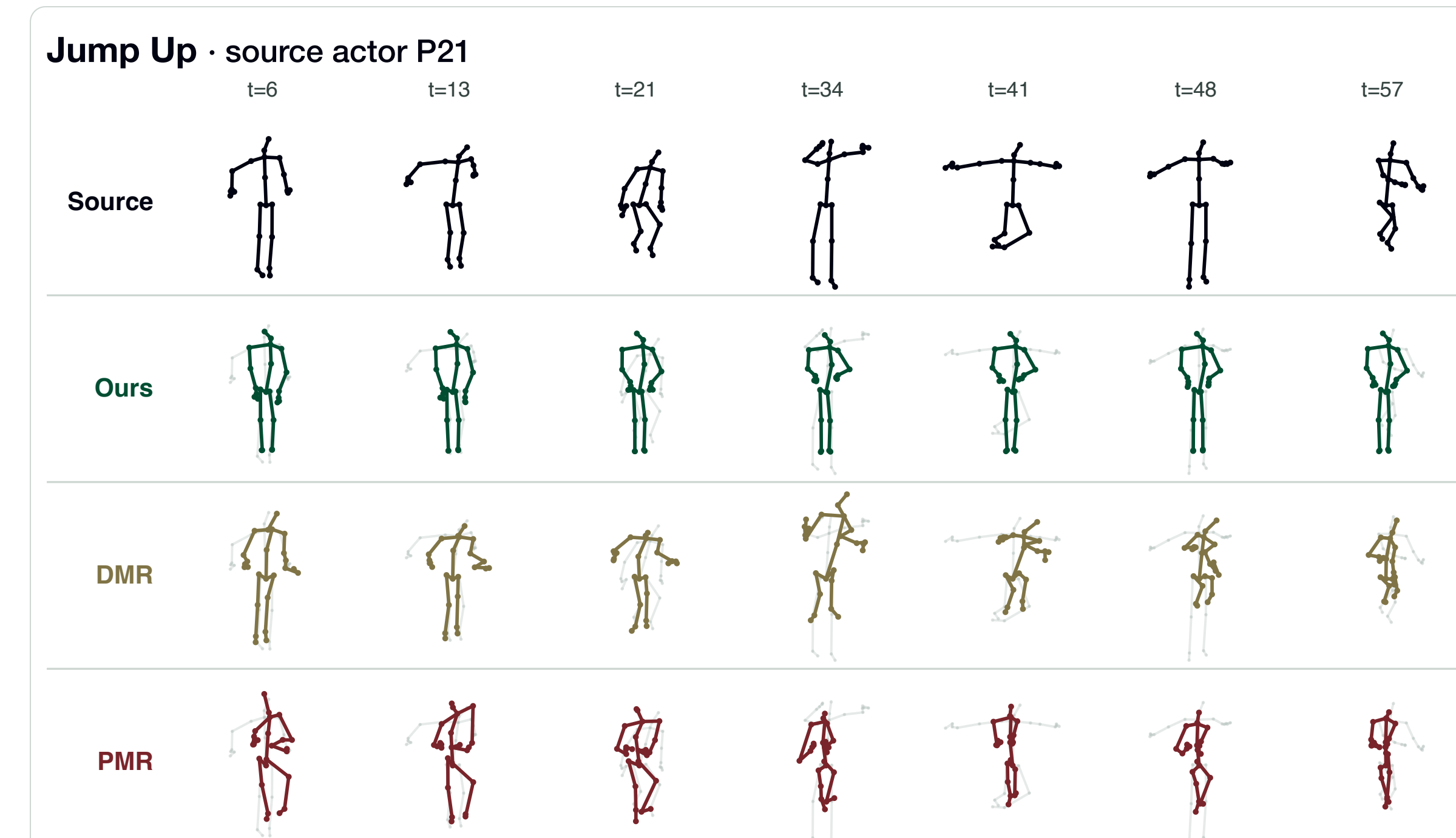
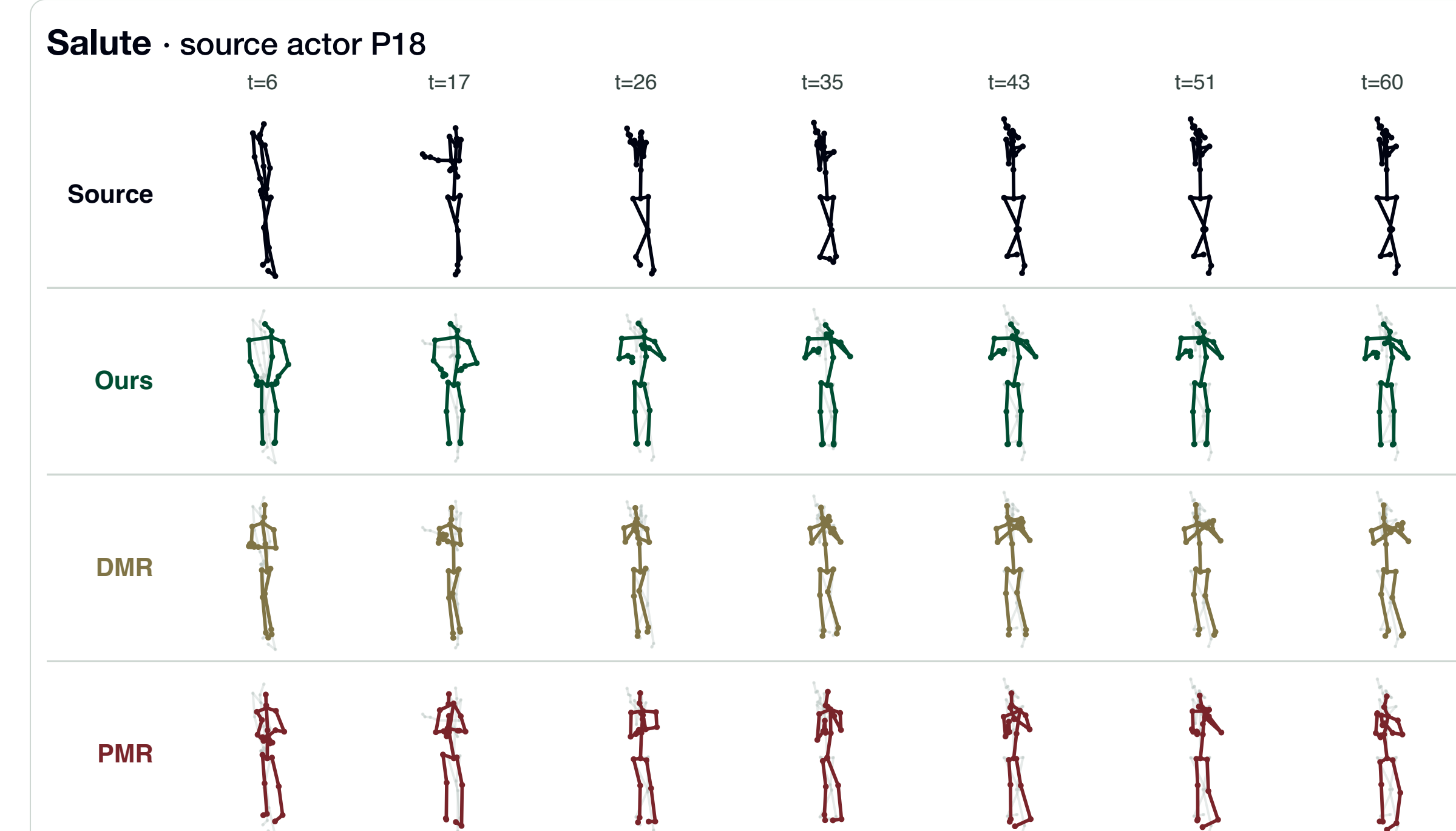


The decoder attends to each stream separately, then blends them with a learned per-channel gate α . Generation is autoregressive under a causal mask.

22.7M parameters, three quarters of them in the decoder, against 4.9M for DMR and 1.0M for PMR.

Retargeting runs once, offline, at collection time, so downstream consumers pay no inference cost. They do retrain on the shared skeletons.

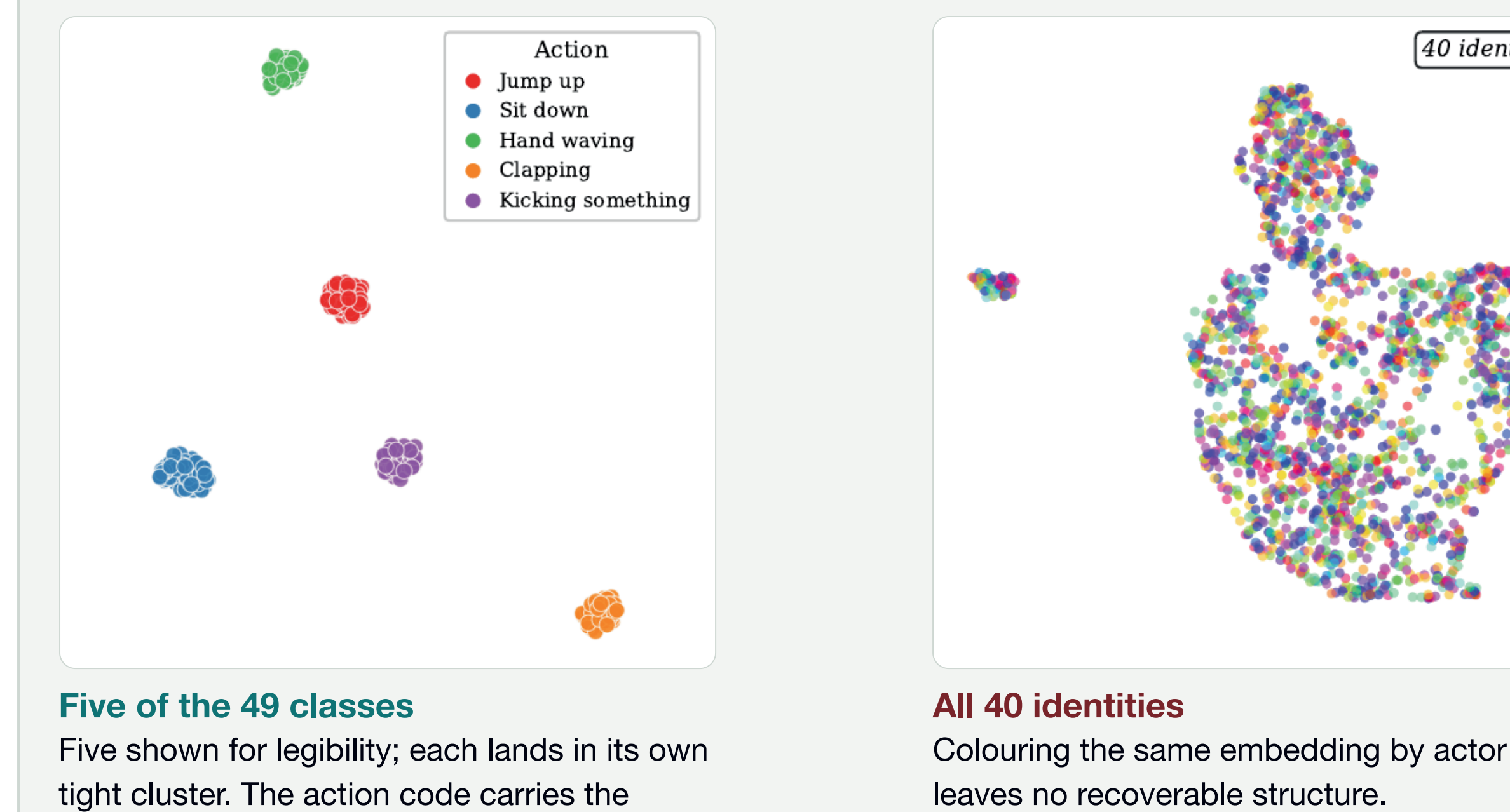
Qualitative results



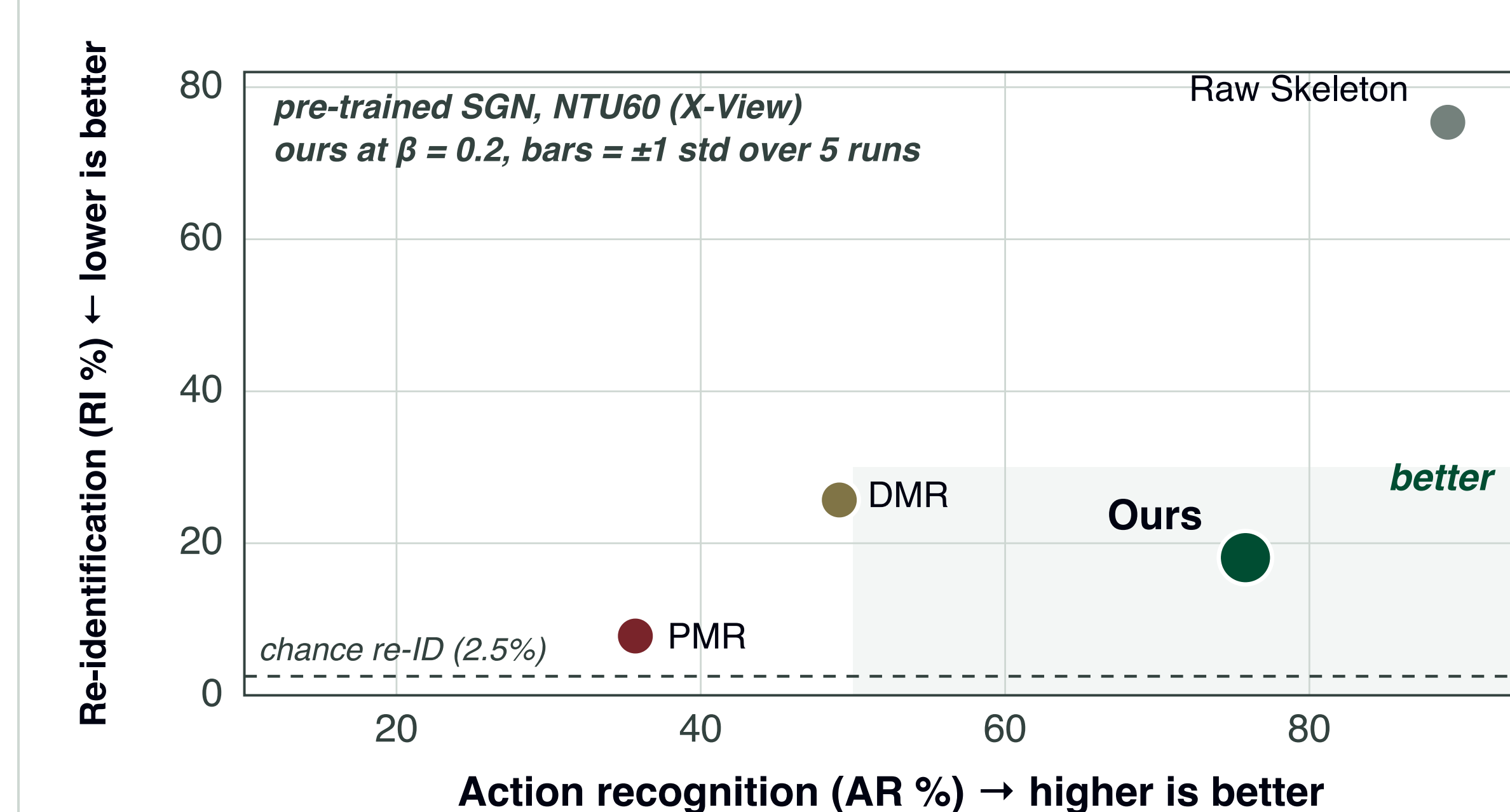
Real sequences from NTU RGB+D 60. **Grey ghost** = source motion; coloured overlay = each method's output. Ours preserves the action on a new body; **DMR** keeps more of the source identity, and **PMR** shifts toward the target but degrades the motion, at 19.9% re-trained AR.

Is it disentangled?

UMAP of the **action embedding**, taken from the action-recognition classifier's features.



Privacy–utility landscape



Down and to the right is better. DMR leaks identity through motion dynamics; PMR buys its low RI by over-anonymising until the action goes with it. Only DisentangledTMR reaches the low-RI regime with the action still readable.

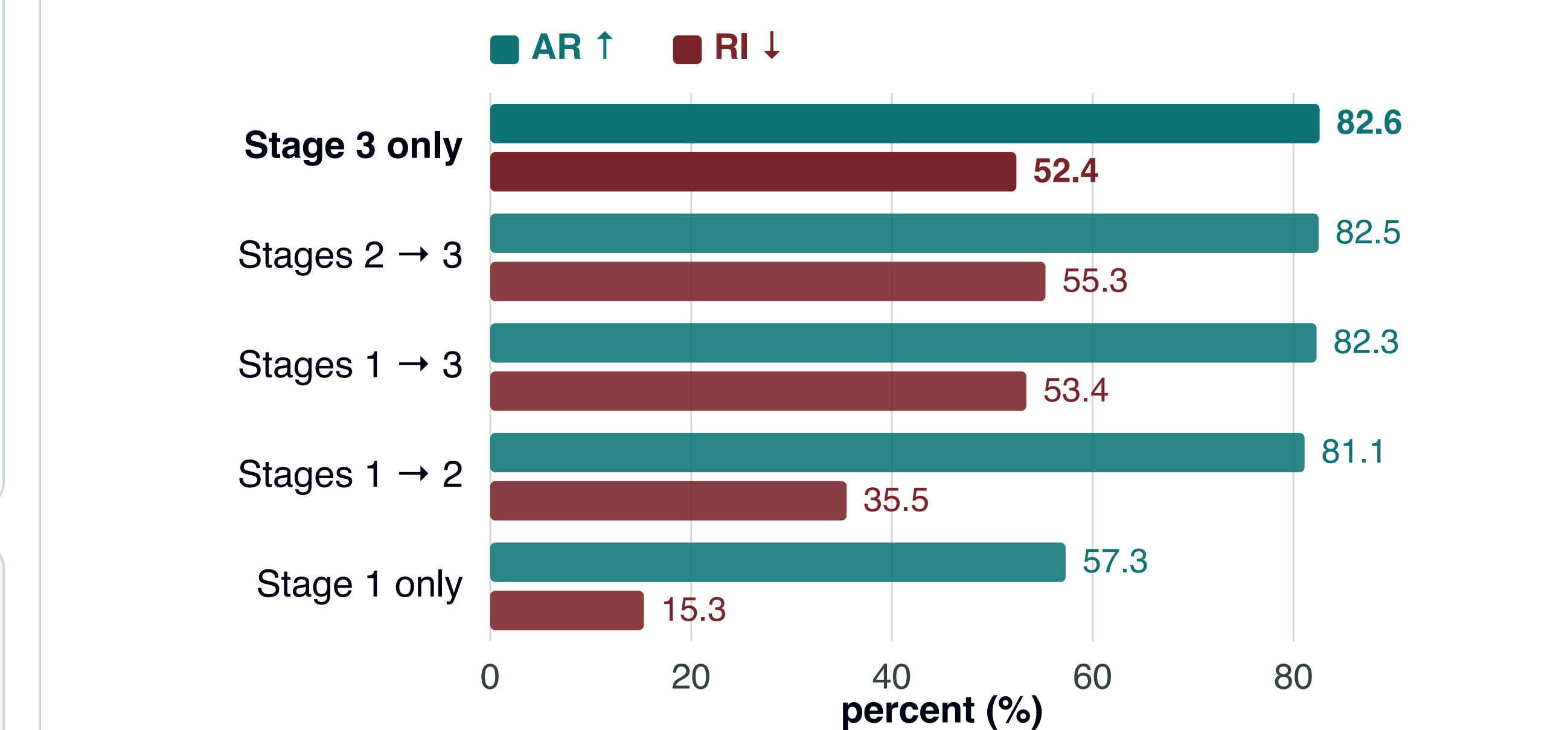
Main results

NTU RGB+D 60, cross-view, ours at $\beta = 0.2$. **Pre-trained** (primary): raw-trained SGN applied directly to retargeted output, no retraining. **Re-trained AR**: a fresh SGN trained on retargeted data. Chance AR $\approx 2\%$, RI $\approx 2.5\%$.

METHOD	PRE-TRAINED SGN		RE-TRAINED	
	AR $\uparrow$	RI $\downarrow$	AR $\uparrow$	RI $\downarrow$
Raw skeleton	89.1	75.4	89.1	75.4
DMR	49.1	25.7	43.1	19.9
PMR	35.7	7.8	19.9	7.8
DisentangledTMR	75.8	18.1	87.1	17.3

Ours is mean $\pm$ std over 5 runs (± 1.1 AR, ± 1.8 RI). **More than double PMR's utility under the same protocol**. PMR reaches a lower RI (7.8%) only by degrading its output until the action is unreadable too, at 19.9% re-trained AR.

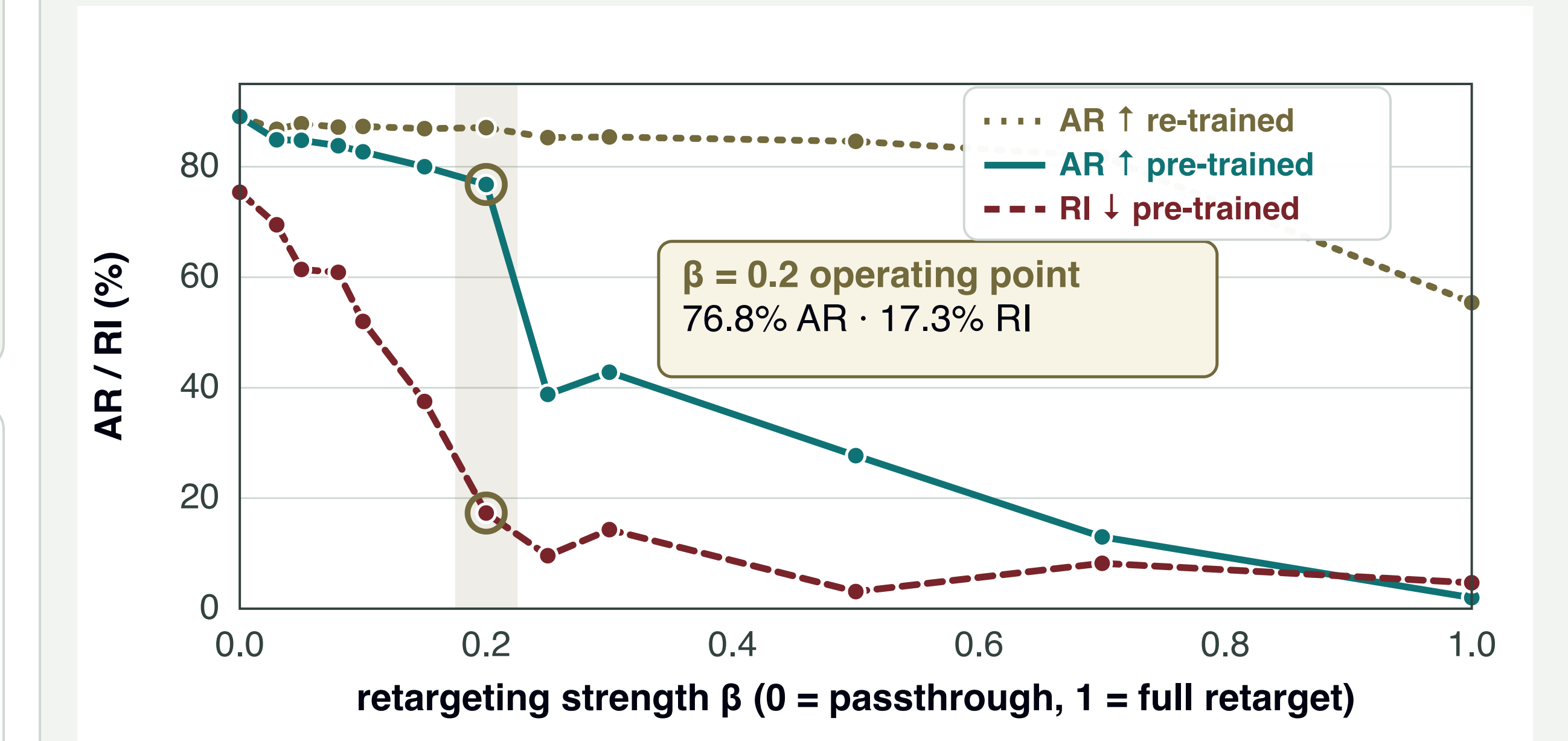
Which stage does the work?



Every configuration that includes **stage 3** lands near 82% AR, so end-to-end fine-tuning does most of the work. **Stage 1 alone** reaches the strongest privacy of any variant, 15.3% RI, but at 57.3% AR the output has lost the structural coherence a classifier needs.

These runs omit the output-level supervision suite, so both columns sit away from the main table: compare rows against each other, not against it.

One knob at test time



β moves the operating point *after* training. A **pre-trained** recogniser reads the action while enough source structure survives, then falls off a cliff between $\beta = 0.20$ and 0.25 , $76.8 \rightarrow 38.8\%$. A **re-trained** one holds about 85% almost the whole way, so the action itself is still there: the cliff is a compatibility limit, not a loss of information. Deployments pick β from their own risk budget.

Takeaways

- Skeleton data is **not** anonymous; treat it as biometric.
- **Replacing** identity beats suppressing it: a single encoder that over-anonymises loses the action too.
- **Asymmetric architecture** does the disentangling; the losses only refine it.
- A single post-hoc β exposes the whole privacy–utility curve from one trained model.